

## SAFETY DATA SHEET

Creation Date 03-May-2010

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** Borane-N,N-diethylaniline complex

**Cat No. :** AC176770000; AC176770250; AC176771000; AC176772500

**Synonyms** N,N-Diethylanilineborane

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |            |
|-------------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                    | Category 4 |
| <b>Acute oral toxicity</b>                                  | Category 3 |
| <b>Acute dermal toxicity</b>                                | Category 3 |
| <b>Acute Inhalation Toxicity</b>                            | Category 3 |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2 |
| Target Organs - Blood.                                      |            |
| <b>Physical Hazards Not Otherwise Classified</b>            | Category 1 |
| Reacts violently with water                                 |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Combustible liquid  
Toxic if swallowed, in contact with skin or if inhaled  
May cause damage to organs through prolonged or repeated exposure  
Reacts violently with water

**Precautionary Statements****Prevention**

Do not allow contact with water  
Keep container tightly closed  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Immediately call a POISON CENTER/doctor  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER/ doctor  
Rinse mouth  
Take off immediately all contaminated clothing  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                                        | CAS-No     | Weight % |
|--------------------------------------------------|------------|----------|
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 13289-97-9 | >95      |
| N,N-Diethylaniline                               | 91-66-7    | <3       |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. May cause methemoglobinemia: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                                                                                                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER                                                                                                    |
| <b>Flash Point</b>                      | 63 °C / 145.4 °F                                                                                                    |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

### Specific Hazards Arising from the Chemical

Reacts violently with water. Contact with water liberates extremely flammable gases. Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Risk of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of boron.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 3                   | 2                  | W                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system.                                                                                                                                                                                            |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition.                                                                                               |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe mist/vapors/spray. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents. Water. Acids. Alcohols. Bases.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations | -               | Splash protection only |
| Neoprene       |                                   |                 |                        |
| Natural rubber |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                     |                               |
|---------------------|-------------------------------|
| Physical State      | Liquid                        |
| Appearance          | Colorless, Amber              |
| Odor                | No information available      |
| Odor Threshold      | No information available      |
| pH                  | No information available      |
| Melting Point/Range | -30 - -27 °C / -22 - -16.6 °F |
| Boiling Point/Range | No information available      |

|                                        |                             |
|----------------------------------------|-----------------------------|
| Flash Point                            | 63 °C / 145.4 °F            |
| Evaporation Rate                       | No information available    |
| Flammability (solid,gas)               | Not applicable              |
| Flammability or explosive limits       |                             |
| Upper                                  | No data available           |
| Lower                                  | No data available           |
| Vapor Pressure                         | No information available    |
| Vapor Density                          | 5.62                        |
| Specific Gravity                       | 0.920                       |
| Solubility                             | Reacts violently with water |
| Partition coefficient; n-octanol/water | No data available           |
| Autoignition Temperature               | No information available    |
| Decomposition Temperature              | No information available    |
| Viscosity                              | No information available    |
| Molecular Formula                      | C10 H18 B N                 |
| Molecular Weight                       | 163.07                      |

## 10. Stability and reactivity

|                                  |                                                                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                         |
| Stability                        | Stable under normal conditions.                                                                                                                             |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Water, Acids, Alcohols, Bases                                                                                                      |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Oxides of boron                                                |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                    |
| Hazardous Reactions              | None under normal processing. Reacts violently with water.                                                                                                  |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                     |
|-------------|-------------------------------------|
| Oral LD50   | Category 3. ATE = 50 - 300 mg/kg.   |
| Dermal LD50 | Category 3. ATE = 200 - 1000 mg/kg. |
| Vapor LC50  | Category 3. ATE = 2 - 10 mg/l.      |

#### Component Information

| Component          | LD50 Oral                | LD50 Dermal         | LC50 Inhalation                           |
|--------------------|--------------------------|---------------------|-------------------------------------------|
| N,N-Diethylaniline | LD50 = 606 mg/kg ( Rat ) | >5000 mg/kg ( Rat ) | LC50 = 1920 mg/m <sup>3</sup> ( Rat ) 4 h |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to respiratory system                                                         |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                               | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Boron,<br>(N,N-diethylbenzenami<br>ne)trihydro-, (T-4)- | 13289-97-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| N,N-Diethylaniline                                      | 91-66-7    | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                                                  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                         |
| <b>Reproductive Effects</b>                       | No information available.                                                                                        |
| <b>Developmental Effects</b>                      | No information available.                                                                                        |
| <b>Teratogenicity</b>                             | No information available.                                                                                        |
| <b>STOT - single exposure</b>                     | None known                                                                                                       |
| <b>STOT - repeated exposure</b>                   | Blood                                                                                                            |
| <b>Aspiration hazard</b>                          | No information available                                                                                         |
| <b>Symptoms / effects, both acute and delayed</b> | May cause methemoglobinemia: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                         |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                   |

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Reacts with water so no ecotoxicity data for the substance is available.

| Component          | Freshwater Algae | Freshwater Fish                                                                                           | Microtox                                          | Water Flea                                            |
|--------------------|------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------|
| N,N-Diethylaniline | Not listed       | LC50: = 16.4 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 38.5 mg/L, 96h (Oncorhynchus mykiss) | EC50 = 6.50 mg/L 5 min<br>EC50 = 7.70 mg/L 15 min | EC50: 1.0 - 1.6 mg/L, 48h semi-static (Daphnia magna) |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Is not likely mobile in the environment.

| Component          | log Pow |
|--------------------|---------|
| N,N-Diethylaniline | 3.31    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                             |                                                                      |
|-----------------------------|----------------------------------------------------------------------|
| <b>UN-No</b>                | UN3148                                                               |
| <b>Proper Shipping Name</b> | consumer commodity Water reactive liquid, n.o.s                      |
| <b>Technical Name</b>       | Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)-, N,N-Diethylaniline |
| <b>Hazard Class</b>         | 4.3                                                                  |
| <b>Packing Group</b>        | I                                                                    |

### TDG

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN3148                       |
| <b>Proper Shipping Name</b> | Water reactive liquid, n.o.s |
| <b>Hazard Class</b>         | 4.3                          |
| <b>Packing Group</b>        | I                            |

### IATA

**UN-No** UN3148  
**Proper Shipping Name** Water reactive liquid, n.o.s  
**Hazard Class** 4.3  
**Packing Group** I

**IMDG/IMO**

**UN-No** UN3148  
**Proper Shipping Name** Water reactive liquid, n.o.s  
**Hazard Class** 4.3  
**Packing Group** I

**15. Regulatory information****International Inventories**

| Component                                        | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 13289-97-9 | -   | X    | X    | ACTIVE                                        | 236-305-2 | -      | -   |
| N,N-Diethylaniline                               | 91-66-7    | X   | -    | X    | ACTIVE                                        | 202-088-8 | -      | -   |

| Component                                        | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 13289-97-9 | -     | -        | -    | X    | X    | -    | -     | -     |
| N,N-Diethylaniline                               | 91-66-7    | X     | KE-10434 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                        | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Boron, (N,N-diethylbenzenamine)trihydro-, (T-4)- | 13289-97-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| N,N-Diethylaniline                               | 91-66-7    | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No | Seveso III Directive (2012/18/EC) - | Seveso III Directive (2012/18/EC) - | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|

|                                                         |            | Qualifying Quantities<br>for Major Accident<br>Notification | Qualifying Quantities<br>for Safety Report<br>Requirements |                |                |
|---------------------------------------------------------|------------|-------------------------------------------------------------|------------------------------------------------------------|----------------|----------------|
| Boron,<br>(N,N-diethylbenzenamine)trihy<br>dro-, (T-4)- | 13289-97-9 | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |
| N,N-Diethylaniline                                      | 91-66-7    | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |

## 16. Other information

**Prepared By**

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**Creation Date**

03-May-2010

**Revision Date**

24-December-2021

**Print Date**

24-December-2021

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**